

Label 1 $\frac{(a^2b^{-3})^2}{a^{-2}b} =$

- A. a^2b^{-7}
- B. a^2b^{-5}
- C. a^6b^{-2}
- D. a^6b^{-6}
- E. a^6b^{-7}

Label 2 $\frac{1}{x+1} + \frac{1}{x-1} + \frac{x+\frac{1}{x}}{x-\frac{1}{x}} =$

- A. $\frac{1}{x+1}$
- B. $\frac{1}{x-1}$
- C. $\frac{x-1}{x+1}$
- D. $\frac{1}{(x+1)(x-1)}$
- E. $\frac{x^2+4x+1}{(x+1)(x-1)}$

Label 3 If $x = \frac{-bx+ay-c}{a+by}$, then $y =$

- A. $\frac{ax+bx+c}{a-bx}$
- B. $\frac{ax+bx+c}{a-bx}$
- C. $\frac{a-bx}{ax+bx+c}$
- D. $\frac{a+bx}{ax+bx+c}$
- E. $\frac{a+bx}{ax-bx-c}$

Label 4 $(2^x)^x =$

- A. $2^{(x^x)}$
- B. 2^x
- C. $2x^x$
- D. 2^{2x}
- E. $2^{(x^2)}$

Label 5 $\left(\frac{\frac{x}{y} + \frac{y}{x} + 2}{\frac{x}{y} - \frac{y}{x}} \right) =$

- A. $\frac{x-y}{x+y}$
- B. $\frac{x+y}{x-y}$
- C. $\frac{x+y}{x-y}$
- D. $\frac{x-y}{x^2+y^2}$
- E. $\frac{x^2-y^2}{x^2+y^2}$

Label 6 If $H = K + \frac{M}{4\pi(r^2+I^2)^n}$ and $r > 0$, then

- A. $\left\{ \left[\frac{M}{4\pi(H-K)} \right]^{-n} \right\}^{\frac{1}{2}}$
Figure 6a Density 316
- B. $\left\{ \left[\frac{M}{4\pi(H-K)} \right]^{\frac{n}{2}} \right\}^{\frac{1}{2}}$
Figure 6b Density 540
- C. $\left\{ \left[\frac{M}{4\pi(H-K)} \right]^{\frac{1}{2}} \right\}^{\frac{1}{2}}$
Figure 6c Density 1080

Figure 6a Density 343

$$\left[\frac{M}{4\pi(H-K)} \right]^{\frac{1}{2n}}$$

Figure 6b Density 1905

$$\left[\frac{M(H-K)}{4\pi} \right]^{\frac{1}{2}}$$

Label 7 If $f(x) = x^2 + x + 1$, then $f(x+1) - f(x) =$

- A. 1
- B. 3
- C. $2x + 1$
- D. $2x + 2$
- E. $x^2 + x + 1$

Label 8 If $\log_{10} x + \log_{10} 4 = \log_{10}(x+4)$, what is the value of x ?

- A. 0
 - B. 1
 - C. 4
 - D. $\frac{1}{3}$
 - E. 4
- x may be any positive number

Label 9 It is given that $x(2x+3) = x(3x-4)$. $x = ?$

- A. 0 only
- B. 7 only
- C. 0 or 7
- D. $-\frac{3}{2}$ or $\frac{4}{3}$ only
- E. $0, -\frac{3}{2}$ or $\frac{4}{3}$

Label 10 $y - 3 > 4y + 2x + 5$ is equivalent to

- A. $y > x + 4$
- B. $y < x + 4$
- C. $y > -x - 4$
- D. $y < -x - 4$
- E. $y > x + 1$

Label 11 The n th term of the arithmetic progression 2, 6, 10, 14, ... is

- A. $2n^2$
- B. $4n$
- C. $4n - 2$
- D. $4n + 2$
- E. $6 - 4n$

Label 12 If $3x - 2y = x + 3y$, then $x^2 : y^2 =$

- A. 2 : 5
- B. 5 : 2
- C. 4 : 25
- D. 25 : 4
- E. 1 : 4

Label 13 The marked price of a book is \$ x . 30% of this price is profit. If the book is sold at a discount of 20%, what will the profit then be?

- A. $\$0.04x$
- B. $\$0.06x$
- C. $\$0.1x$
- D. $\$0.24x$
- E. $\$0.56x$

Label 14 A group consists of n boys and n girls. If two of the girls are exceeded by two other boys, then 51% of the group members will be boys. What is n ?

- A. 50
- B. 51
- C. 52
- D. 100
- E. 102

Label 15 If the surface area of a spherical soap bubble increases by 44%, its volume increases by

- A. 20%
- B. 33.1%
- C. 60%
- D. 66%
- E. 72.8%

Label 16

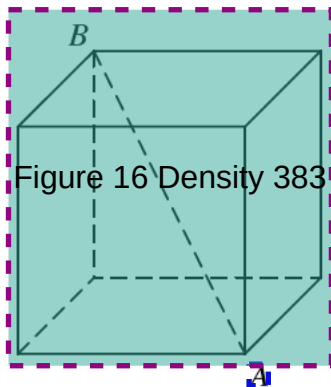


Figure 16 Density 383

The total area of the six faces of the solid cube in the figure is 96 cm^2 . What is the length of the diagonal AB ?

- A. $6\sqrt{2} \text{ cm}$
- B. $4\sqrt{3} \text{ cm}$
- C. $4\sqrt{2} \text{ cm}$
- D. $2\sqrt{6} \text{ cm}$
- E. 4 cm

Label 17 A merchant sold 100 chairs. 80 of them were sold at a profit of 30% on each chair, while 20 of them were sold at a loss of 40% on each chair. What is his percentage gain or loss on the whole stock?

- A. A loss of 8%
- B. A loss of 10%
- C. A gain of 8%
- D. A gain of 16%
- E. A gain of 24%

Label 18 If $0^\circ < \theta < 90^\circ$ and $\sin \theta = \frac{k}{2}$, then $\cos \theta =$

- A. $1 - \frac{k}{2}$
- B. $\frac{2}{\sqrt{4+k^2}}$
- C. $\frac{\sqrt{4+k^2}}{2}$

Figure 18 Density 1745

$$\frac{\frac{2}{\sqrt{4+k^2}}}{\frac{\sqrt{4-k^2}}{2}}$$

Label 19 $\tan \theta \sin \theta - \frac{1}{\cos \theta} =$

- A. 0
- B. $\cos \theta$
- C. $-\cos \theta$
- D. $\frac{-\cos \theta}{\cos \theta}$
- E. $-\tan \theta \sin \theta$

Figure 19 Density 4711

Label 20 If $0^\circ \leq \theta \leq 360^\circ$, the number of roots of the equation $2 \sin \theta \cos \theta - \cos \theta = 0$ is

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

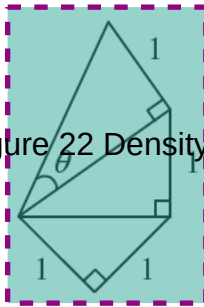
Label 21 An angle measures x radians. What is its measure in degrees?

- A. $\left(\frac{\pi x}{180}\right)^\circ$
- B. $\left(\frac{180x}{\pi}\right)^\circ$
- C. $\left(\frac{\pi}{180x}\right)^\circ$
- D. $\left(\frac{\pi}{360}\right)^\circ$
- E. $\left(\frac{360x}{\pi}\right)^\circ$

Figure 21 Density 348

Label 22

Figure 22 Density 72

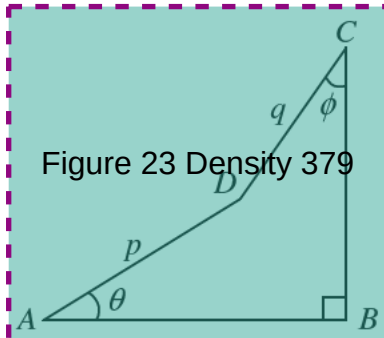


In the figure, $\cos \theta =$

- A. $\frac{1}{2}$
- B. $\frac{2}{3}$
- C. $\frac{3}{4}$
- D. $\frac{\sqrt{3}}{2}$
- E. $\frac{\sqrt{3}}{4}$

Label 23

Figure 23 Density 379

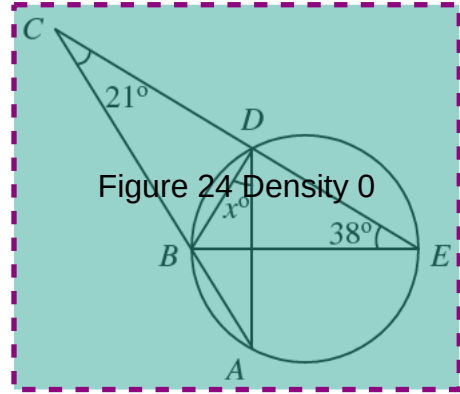


In the figure, $AD = p$, $DC = q$, $\angle B = 90^\circ$. $AB =$

- A. $p \sin \theta + q \sin \phi$
- B. $p \cos \theta + q \cos \phi$
- C. $p \sin \theta + q \cos \phi$
- D. $p \cos \theta + q \sin \phi$
- E. $(p + q)(\cos \theta + \cos \phi)$

Label 24

Figure 24 Density 0

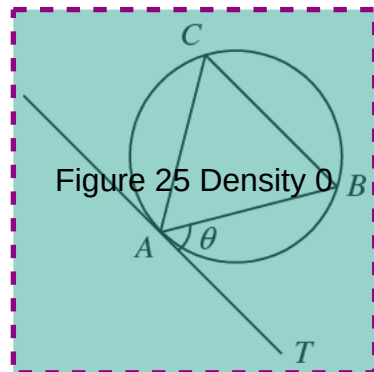


In the figure, BE is a diameter of the circle. ABC and EDC are straight lines. $x^\circ =$

- A. 21°
- B. 31°
- C. 38°
- D. 52°
- E. 59°

Label 25

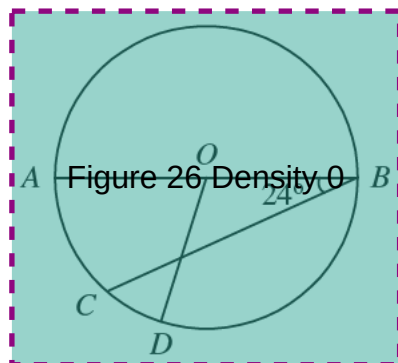
Figure 25 Density 0



In the figure, AT touches the circle at A . In $\triangle ABC$, $\angle A : \angle B : \angle C = 2 : 3 : 4$. $\theta =$

- A. 40°
- B. 50°
- C. 60°
- D. 70°
- E. 80°

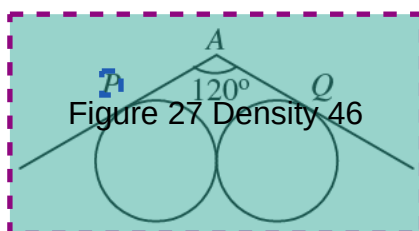
Label 26



In the figure, AB is a diameter of the circle with centre at O . The length of the minor arc AC is twice the length of the minor arc CD . $\angle BOD =$

- A. 72°
- B. 90°
- C. 108°
- D. 132°
- E. 144°

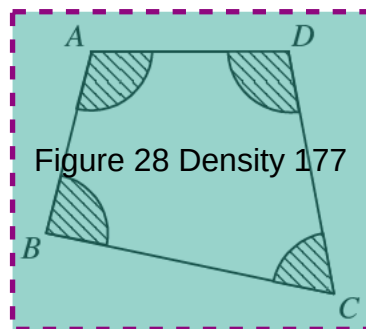
Label 27



In the figure, two circles both with radius 2 cm touch each other externally. AP and AQ are equal tangents to the two circles. $AP = ?$

- A. $\sqrt{3}$ cm
- B. $2\sqrt{3}$ cm
- C. 4 cm
- D. $4\sqrt{3}$ cm
- E. $\frac{4\sqrt{3}}{3}$ cm

Label 28



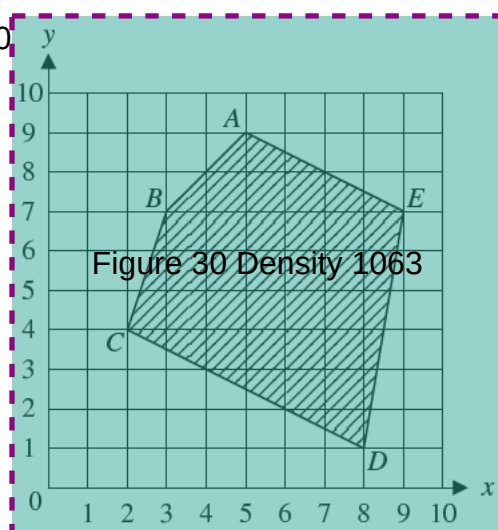
In the figure, $ABCD$ is a quadrilateral. The shaded portions are four sectors with centres at A, B, C and D . Their radii are all equal to a . What is the total area of the four sectors?

- A. πa^2
- B. $2\pi a^2$
- C. $4\pi a^2$
- D. $\sqrt{2} \pi a^2$
- E. It cannot be determined

Label 29 $x^2 - 2 \leq 0$ is equivalent to

- A. $x \leq 1$
- B. $x \geq -1$
- C. $-1 \leq x \leq 1$
- D. $x \geq 1$ or $x \leq -1$
- E. $x \leq 1$ or $x \geq -1$

Label 30



In the figure, which point in the shaded region will make the value of $x - 2y$ a minimum.

- A. A
- B. B
- C. C
- D. D
- E. E

Label 31 $6x^2 + kx + 6 = 0$ is a quadratic equation in which k is a constant. Its roots α and β are positive. $\log_{10}\alpha + \log_{10}\beta =$

- A. 0
- B. 1
- C. $\log_{10}6$
- D. $\log_{10}(-k)$
- E. $\log_{10}\left(-\frac{k}{6}\right)$

Label 32 $-3x^2 - 3x \equiv -3(x + a)^2 + b$ is an identity in x . What are the values of the constants a and b ?

- A. $a = 1$ and $b = 0$
- B. $a = \frac{1}{2}$ and $b = \frac{3}{4}$
- C. $a = \frac{1}{2}$ and $b = \frac{3}{4}$
- D. $a = -\frac{1}{2}$ and $b = \frac{3}{4}$
- E. $a = -\frac{1}{2}$ and $b = -\frac{3}{4}$

Label 33 The H.C.F. and L.C.M. of three expressions are a^2b^2c and $a^4b^6c^4$ respectively. Two of the expressions are $a^2b^3c^4$ and $a^3b^2c^2$. The third expression is

- A. a^3b^3c
- B. $a^3b^6c^4$
- C. a^4b^2c
- D. a^4b^6c
- E. $a^4b^6c^2$

Label 34 The sum of the first five terms of an arithmetic progression is 15. If the fourth term is 7, the first term is

- A. -5
- B. -3
- C. -1
- D. 1
- E. 10

Label 35 Which of the following can be summed to infinity?

- I. The arithmetic progression 4, 3, 2, 1,
- II. The geometric progression 27, 9, 3, 1,
- III. The geometric progression 16, -8, 4, -2,

- A. II only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

Label 36 The running speeds of three boys A, B and C are in the ratios $a : b : c$. The times that A, B and C take to complete a 1500 m race are in the ratios

- A. $a : b : c$
- B. $c : b : a$
- C. $b + c : a + c : a + b$
- D. $\frac{a}{b} : \frac{b}{c} : \frac{c}{a}$
- E. $\frac{a}{b} : \frac{b}{c} : \frac{c}{a}$

Label 37 If n is a positive integer, which of the following numbers is/are odd?

- I. 2^{2n+1}
- II. $3(2^n)$
- III. $(2n+1)^2$

- A. II only
- B. III only
- C. I and III only
- D. II and III only
- E. I, II and III

Label 38
38. A factory employs x workers each working n hours a day. The whole factory produces k watches per day. If y workers go on leave, then how many hours a day should the remaining workers work in order to produce the same number of watches per day?

- A. $\frac{nx}{y}$
 B. $\frac{ny}{x}$
 C. $\frac{nx}{4y}$
 D. $\frac{nx}{x-y}$
 E. $\frac{n(x-y)}{x}$

Label 39
39. The daily wages of a man and a boy are in the ratio 2 : 1. In a day a man has to work 8 hours but a boy only 6 hours. The hourly wages of a man and a boy are in the ratio

- A. 8 : 3
 B. 2 : 1
 C. 3 : 2
 D. 4 : 3
 E. 1 : 1

Label 40
40.

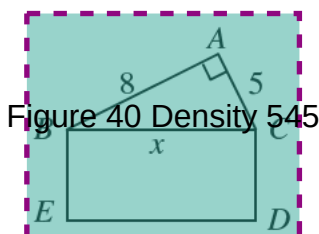


Figure 40 Density 545

In the figure, $\angle BAC = 90^\circ$, $AB = 8$, $AC = 5$ and $AX \perp BC$. $BCDE$ is a rectangle with $CD = AX$. What is the area of the rectangle $BCDE$?

- A. 20
 B. 40
 C. 80
 D. 89

E. $4\sqrt{89}$

Label 41
41.

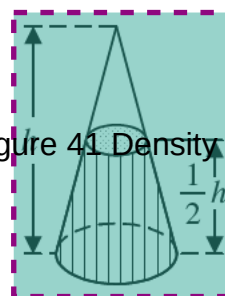


Figure 41 Density 334

The height of the cone in the figure is h . It contains water to a depth of $\frac{1}{2}h$.

$$\frac{\text{Volume of water}}{\text{Capacity of the cone}} =$$

- A. $\frac{1}{8}$
 B. $\frac{1}{4}$
 C. $\frac{1}{2}$
 D. $\frac{3}{4}$
 E. $\frac{7}{8}$

Label 42
42.

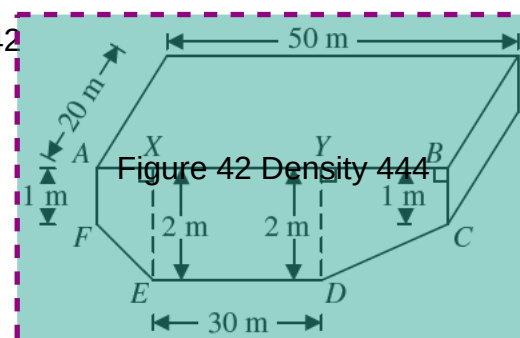


Figure 42 Density 444

The figure above represents a $50\text{m} \times 20\text{m}$ swimming pool. The pool is in the shape of a prism with a rectangular surface and four vertical walls. The dimensions of the sidewall $ABCDEF$ are as shown in the figure. What is the capacity of the pool in m^3 ?

- A. 1200
 B. 1500

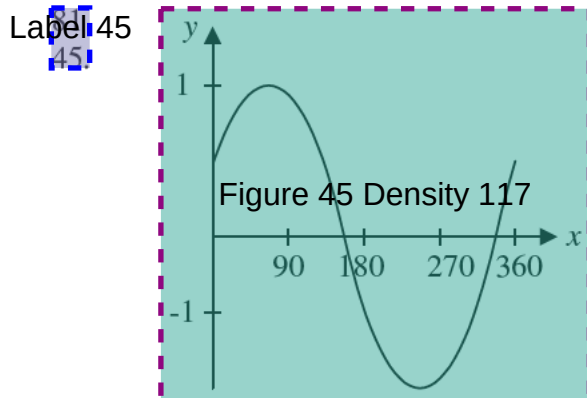
- C. 1800
D. Figure 42 Density 3385
E. It cannot be determined

Label 43
43. Given that $\sin \theta - \cos \theta = \frac{1}{2}$, what is the value of $\sin \theta \cos \theta$?

- A. $\frac{1}{2}$
B. $\frac{1}{4}$
C. $\frac{3}{8}$
D. $\frac{3}{4}$
E. It cannot be determined

Label 44
44. If $0^\circ \leq \theta \leq 360^\circ$, the minimum value of $1 + 2\cos \frac{\theta}{2}$ is

- A. -2
B. -1
C. 0
D. 1
E. 2



The figure above shows the graph of

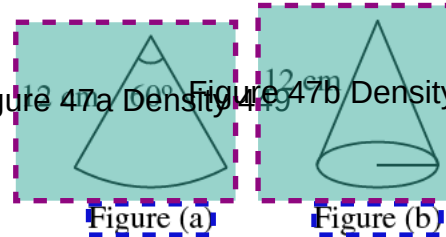
- A. $y = \sin(x^\circ + 30^\circ)$
B. $y = \sin(x^\circ - 30^\circ)$
C. $y = \sin(x^\circ + 150^\circ)$
D. $y = \sin(x^\circ - 150^\circ)$
E. $y = \sin(x^\circ + 60^\circ)$

Label 46
46. The radius of a sector is 3 cm and the perimeter is 10 cm. What is the area of the sector?

- A. 6 cm^2
B. 12 cm^2
C. 15 cm^2
D. 18 cm^2
E. 45 cm^2

Label 47
47.

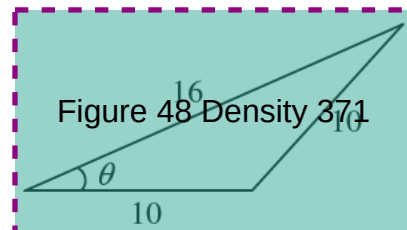
Figure 47a Density 478
Figure 47b Density 472



The cone in Figure (b) is formed by bending the sector in Figure (a). The angle of the sector is 60° and the radius is 12 cm. The radius of the base of the cone is

- A. 2 cm
B. 4 cm
C. 6 cm
D. $2\pi \text{ cm}$
E. $\frac{360}{\pi} \text{ cm}$

Label 48
48.



In the figure, $\sin \theta =$

- A. 0.5
B. 0.6
C. 0.625
D. 0.75
E. 0.8

Label 49

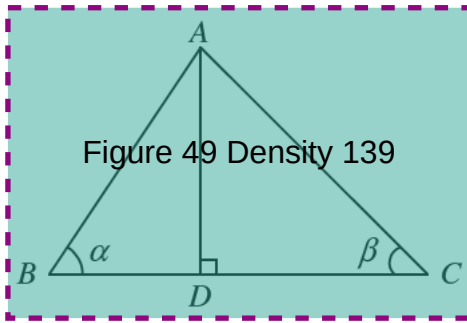


Figure 49 Density 139

In the figure, $AD \perp BC$, $CD =$

- A. $h \sin \alpha \tan \beta$
- B. $h \cos \alpha \tan \beta$
- C. $h \tan \alpha \sin \beta$
- D. $\frac{h \cos \alpha}{\tan \beta}$
- E. $\frac{h \sin \alpha}{\tan \beta}$

Label 50

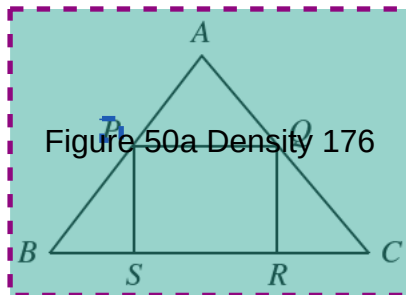


Figure 50a Density 176

In the figure, ABC is an equilateral triangle of side $2a$. P and Q are the mid-points of AB and AC respectively. $PQRS$ is a rectangle. What is the area of $PQRS$?

- A. a^2
- B. $\frac{1}{2}a^2$
- C. $\frac{2}{3}a^2$
- D. $\frac{1}{\sqrt{3}}a^2$
- E. $\frac{\sqrt{3}}{2}a^2$

Figure 50b Density 2806

Label 51

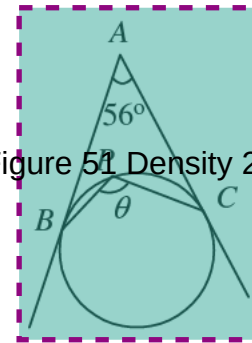


Figure 51 Density 207

In the figure, AB and AC touch the circle at B and C . If P is any point on the minor arc BC , what is θ ?

- A. 112°
- B. 118°
- C. 124°
- D. 146°
- E. It cannot be determined

Label 52

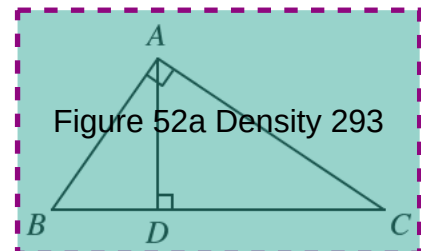


Figure 52a Density 293

$\angle BAC = 90^\circ$
 $AD \perp BC$

11

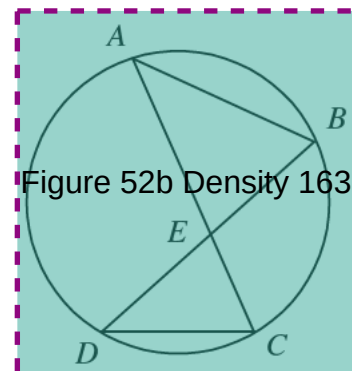
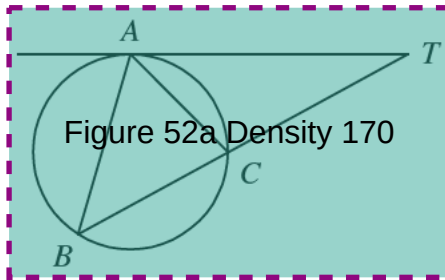


Figure 52b Density 163

AC and BC
intersect at E

III



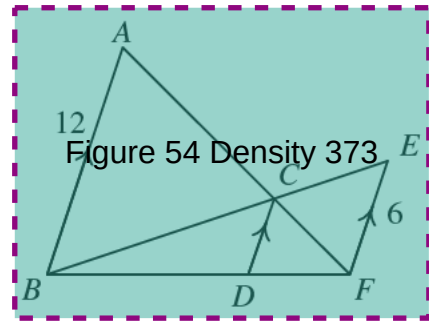
BC produced meets the tangent AT at T .

Which of the above figures contains one or more pairs of similar triangles?

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

Figure 52b Density 5318

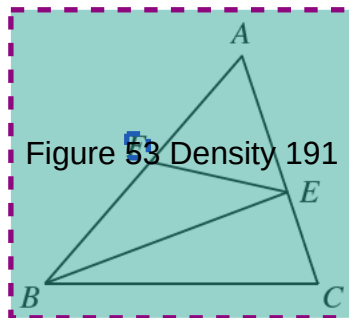
Label 54



In the figure, $AB \parallel CD \parallel EF$, ACE , BCE and BDF are straight lines. $AB = 12$, $EF = 6$. $CD = ?$

- A. 4.5
- B. 4
- C. 3.6
- D. 3
- E. 2

Label 53



In the figure, P is the mid-point of AB , E is a point on AC such that $AE : EC = 2 : 1$. $\frac{\text{Area of } \triangle BFE}{\text{Area of } \triangle BCE} =$

- A. 1
- B. 2
- C. 3
- D. 1
- E. 2